



Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division

Course: AIML ZG528 – AI and ML for Robotics
Assignment 2: Implementation of Phases in Robotics

Modules Covered Module #1- Module #5(Till Graph Based Planners only)

Total Marks: 20

Deadline: As announced on eLearn

Type: Work-Integrated Learning Assignment (Part A - Research Paper Study, Part B -Python Implementation)

Submission: PDF report (3–4 pages **only**) + Python code notebook with screenshots

Learning Objectives

After completing this assignment, students will be able to:

- Apply robotics concepts to realistic workplace-inspired problems.
- Evaluate how robot phases perform using AI&ML technique.
- Design & implement localization, mapping & planning inROS2.

Part A – Study

Research Paper: To access the assigned paper, [click here with your BITS EMAIL ONLY](#). Review the Phases of Robotic Design in the given research paper and identify below pointers: Refer to the live contact session for further clarity on the below requirement.

- Objective
- Domain
- Scope of the Work
- Key area open for future enhancement and/or Optimization
- List precisely key design aspect w.r.t following phases of mobile robotics (Important Note: Brief each of the below in no more than 3 sentences)
 - Deployment Platform
 - Sensors & Types
 - Perception: Sensor Measurement Model & Motion Model
 - Localization & Mapping
 - Trajectory Planning
 - Control Technique
 - Communication Infrastructure (IoT/Cloud/Wifi etc.,)

Part B – Implementation

Scenario: You are working in a team deploying autonomous robots in a workplace setting. **Choose the same domain based on your group's preference chosen during assignment 1 from below:**

- Inventory bots (movement, management, monitoring etc.,)
- Warehouse bots (logistics, inspection etc.,)
- Healthcare bots (delivery, monitoring, assistance etc.,)
- Hospitality bots (delivery, monitoring, assistance etc.,)
- Manufacturing bots (process inspection, material handling, assembly etc.,)
- Agricultural bots (inspection, material handling, delivery, harvesting etc.,)
- Environmental field bots (monitoring, inspection, deployment, rescue, clearing, sample collection etc.,)
- Aviation bots (assistance, search, combat, logistics etc.,)
- Software bots (any automation agents where all the phases of robotics must be applicable)



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Implement all the following algorithms for the chosen domain with a simulation tool of your choice. Ensure that all the group members consent & are ready to work in your group preferred simulation tool of choice.

- Monte Carlo Localization
- Grid based SLAM
- D* Planner
- Apply Reinforcement learning to complement any of the sub activities in above traditional SLAM in part B). Justify your choice & relevance to the algorithm.
- Replace the above traditional planner pipeline in part C), with traditional deep learning models like CNN or RNN. Justify your choice & relevance to the algorithm.

Submission Format (Applicable only for Part A questions)

- File 1: Assignment Report (PDF, max 4 pages)
- File 2: Code notebook (.ipynb) . Mention the BITS ID and NAME of the group members who contributed to the assignment exercises.
- Screenshots/ Simulation output copy of the codes can be appended in the File 1 PDF report.

Tools: Python, Google Colab, PyBullet or Gazebo/RVIZ.

Deadline: As announced on eLearn

Marking Scheme

The grading will be relative based on the class performance. **Key factors** which will be considered for scoring includes (but not limited to): Creativity, Originality, Uniqueness, Relevance to the chosen domain's open problems, Code Modularity, Representation, Reproducibility, Similarity to AI tool solutions.

Component	Description	Marks
Part A – Research Paper Study	Components & Phases of Robotics - Design , Modelling & Implementation, Comparative Analysis to traditional techniques	2
Part B – Python Implementation	Ability to apply, relate to Assignment 1 chosen domain, implement using python programming, infer the outcome , critically evaluate the performance	8
Part B – Python Implementation	Usage of the BITS Virtual Cloud Labs, Implementation in ROS2(at least one part (not limited to) of the above assignment subparts must be in ROS2)	2
Viva(Individual)	Ability to demonstrate implementation skills in all the above. Irrespective of individual contribution a student may be examined on any parts of the assignment done by the group	8
Total		20